

9.13 Solution: By the same consideration as in Prob. 9.12, the magnetization density in the sphere is

$$\mathbf{m} = \frac{M}{4\pi R_0^3/3} \hat{\mathbf{z}},$$

to the lowest order in β . Since there is no electric charge and current, and also

$$\nabla \cdot (\mathbf{r} \times \mathbf{m}) = (\nabla \times \mathbf{r}) \cdot \mathbf{m} - \mathbf{r} \cdot (\nabla \times \mathbf{m}) = 0,$$

by Eq. (9.170), all electric multipole moments are zero, and we only need to consider the magnetic moments M'_{lm} in Eq. (9.172). The magnetization density is discontinuous at the surface of the sphere and there will be non-zero surface magnetization density, given by $\mathbf{m} \cdot \mathbf{n} = m \cos \theta$. Therefore,

$$M'_{lm} = - \int r^l Y_{lm}^*(\theta, \phi) m \cos \theta \delta(r - R(\theta)) d^3x.$$

Clearly, only $m = 0$ moment is non-zero. With

$$Y_{lm}(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi}} P_l(\cos \theta),$$

we have

$$M'_{l0} = -\sqrt{\frac{2l+1}{4\pi}} 2\pi m R_0^{l+2} \int_{-1}^1 (1 + \beta P_2(\cos \theta))^{l+2} P_l(\cos \theta) \cos \theta d(\cos \theta).$$

To the lowest order in β , when $l = 0$, the moment is

$$M'_{00} = -\sqrt{\frac{1}{4\pi}} 2\pi m R_0^2 \int_{-1}^1 2\beta P_2(\cos \theta) \cos \theta d(\cos \theta) = 0.$$

For $l = 1$,

$$\begin{aligned} M'_{10} &= -\sqrt{\frac{3}{4\pi}} 2\pi m R_0^3 \int_{-1}^1 3\beta P_2(\cos \theta) P_1(\cos \theta) \cos \theta d(\cos \theta) \\ &= -\sqrt{\frac{3}{4\pi}} 6\pi m R_0^3 \beta \int_{-1}^1 \frac{1}{2} (3x^2 - 1) x^2 dx \\ &= -\sqrt{\frac{3}{4\pi}} 3\pi m R_0^3 \beta \cdot \frac{8}{15} = -\frac{6}{5} \sqrt{\frac{3}{4\pi}} M \beta. \end{aligned}$$

Using Eq. (9.151), the power radiated per unit solid angle is

$$\begin{aligned} \frac{dP(1,0)}{d\Omega} &= \frac{Z_0}{2k^2} |a_M(1,0)|^2 |\mathbf{X}_{10}|^2 = \frac{Z_0}{2k^2} \left| \frac{ik^3}{3} \sqrt{2} M'_{10} \right|^2 |\mathbf{X}_{10}|^2 \\ &= \frac{Z_0}{2k^2} \frac{2k^6}{9} \frac{3}{4\pi} \frac{36}{25} M^2 \beta_0^2 \frac{3}{8\pi} \sin^2 \theta = \frac{9Z_0 k^4}{200\pi^2} M \beta_0^2 \sin^2 \theta, \end{aligned} \quad (1)$$

and the total power radiated is

$$P(1,0) = \frac{Z_0}{2k^2} |a_M(1,0)|^2 = \frac{Z_0}{2k^2} \frac{2k^6}{9} \frac{3}{4\pi} \frac{36}{25} M^2 \beta_0^2 = \frac{3Z_0 k^4}{25\pi^2} M^2 \beta_0^2.$$